

Assignment 1: Stage I Planning

Background of the project

Company XYZ needs to start using a database and is having multiple challenges with managing the data within its organization. Company XYZ received a referral from your university professor and shared that your database consulting firm is thriving and that, with your superior database management skills and business acumen, you would be an ideal candidate to build a database for their organization. You have been asked to fully design a mock database using Microsoft Access that could be implemented at this organization. They are currently vetting other consulting firms and would like to give your firm the \$1.5M contract. Using the Database Development Life Cycle, your consulting firm will develop a fully working database demonstrating various requirements. You have complete autonomy and creative freedom to determine the situation and business needs. The aim of the assignment and presentation is to assess and evaluate the following:

- a) Ability to use the lexicon (language) when discussing databases.
- b) Ability to briefly discuss and demonstrate complex technical concepts (i.e., a database), their advantages, and aspects to a layman, creatively and succinctly.
- c) Ability to follow (at a high level) the Database Development Life Cycle to design and develop a database meeting requirement.

Assignment Details

1. Determine the name of your Consulting Firm:

Carter Consulting Company (CCC)

2. Determine the organization name (Company XYZ) for which you will develop the database.

SwiftShip Deliveries (SSD)

3. Provide a brief paragraph about this organization (what they do, why they do it, etc.).

SwiftShip Deliveries is a company that aims to provide fast and reliable shipment services for both businesses and individuals. SwiftShip focuses on ensuring that deliveries reach their destination swiftly and safely.

4. Describe the benefit of developing this database using MS Access.

Using Microsoft Access to create this database will have multiple advantages. Access provides a very user-friendly interface for both designing and managing databases. Access also allows for easy retrieval and analysis of data. Compared to using an Excel spreadsheet, Access offers data integrity and security features, and can be scaled easily, making it easier to maintain the database as the company grows.

5. Determine the following information regarding the initial plans of your database:

- a) What information will exist within your database?
SwiftShip Deliveries would like to have a database that keeps track of driver safety. The database will keep track of drivers, managers, incidents, and various types of violations, such as speeding, seatbelt usage, harsh braking, etc.
- b) Using the format below, complete the information below to discuss the tables used within your database and the related fields, data types, and descriptions. For example, if you use five tables, you will complete the information below for each table.

Table Name: ***Drivers***

Which field is the primary key (place an X)	Fields	Data Type	Field Description
<i>X</i>	<i>DriverID</i>	<i>Short Text</i>	<i>Unique identification for each driver</i>
	<i>LastName</i>	<i>Short Text</i>	<i>Last name of the driver</i>
	<i>FirstName</i>	<i>Short Text</i>	<i>First name of the driver</i>
	<i>Email</i>	<i>Short Text</i>	<i>Email used to contact the driver</i>
	<i>Phone</i>	<i>Short Text</i>	<i>Phone number used to contact the driver</i>
	<i>Street</i>	<i>Short Text</i>	<i>Street address where the driver lives</i>
	<i>City</i>	<i>Short Text</i>	<i>City where the driver lives</i>
	<i>State</i>	<i>Short Text</i>	<i>State where the driver lives</i>
	<i>ZipCode</i>	<i>Short Text</i>	<i>ZIP code of the area where the driver lives</i>
	<i>HireDate</i>	<i>Date/Time</i>	<i>Date that the driver started working for the company</i>
	<i>TotalMileage</i>	<i>Number</i>	<i>Total mileage the driver has driven while working for the company</i>
	<i>ManagerID</i>	<i>Short Text</i>	<i>Foreign key linking to the Managers Table to identify the driver's manager</i>

Table Name: *Managers*

Which field is the primary key (place an X)	Fields	Data Type	Field Description
<i>X</i>	<i>ManagerID</i>	<i>Short Text</i>	<i>Unique identification for each manager</i>
	<i>LastName</i>	<i>Short Text</i>	<i>Last name of the manager</i>
	<i>FirstName</i>	<i>Short Text</i>	<i>First name of the manager</i>
	<i>Email</i>	<i>Short Text</i>	<i>Email used to contact the manager</i>
	<i>Phone</i>	<i>Short Text</i>	<i>Phone number used to contact the manager</i>
	<i>Street</i>	<i>Short Text</i>	<i>Street address where the manager lives</i>
	<i>City</i>	<i>Short Text</i>	<i>City where the manager lives</i>
	<i>State</i>	<i>Short Text</i>	<i>State where the manager lives</i>
	<i>ZipCode</i>	<i>Short Text</i>	<i>ZIP code of the area where the manager lives</i>

Table Name: *Incidents*

Which field is the primary key (place an X)	Fields	Data Type	Field Description
<i>X</i>	<i>IncidentID</i>	<i>Short Text</i>	<i>Unique identification for each incident record</i>
	<i>DriverID</i>	<i>Short Text</i>	<i>Foreign key linking to the Drivers Table to identify the driver</i>
	<i>ViolationID</i>	<i>Short Text</i>	<i>Foreign key linking to the Violations table to identify the type of violation</i>
	<i>IncidentDate</i>	<i>Date/Time</i>	<i>Date that the incident occurred</i>
	<i>IncidentTime</i>	<i>Date/Time</i>	<i>Time that the incident occurred</i>

Table Name: ***Violations***

Which field is the primary key (place an X)	Fields	Data Type	Field Description
<i>X</i>	<i>ViolationID</i>	<i>Short Text</i>	<i>Unique identification for each type of violation</i>
	<i>ViolationType</i>	<i>Short Text</i>	<i>Short description of the type of violation</i>